

Spotify 2023 Data Analysis using Power BI

Music Industry Analyst Report

Dataset: spotify-2023.csv (952 songs analysed after cleaning)

Tool: Microsoft Power BI Desktop (Power Query and DAX)

1. Introduction

This report presents an analysis of the Spotify 2023 dataset from the viewpoint of an analyst in the music industry. The aim is to identify patterns and relationships in songs, platforms, playlists, charts and artists that can support decision-making. The analysis covers six tasks: audio features, cross-platform presence, playlist exposure and streams, chart position and streams, outlier songs, and artist concentration (80/20 analysis). All analysis was carried out in Power BI. Forecasting and predictive modelling were not part of the scope.

Throughout the report, findings are separated into **evidence from the data** and **possible interpretations**. Relationships are described as association or correlation. No causal claims are made.

2. Data and preparation

2.1 Dataset

The imported file has 953 rows and 24 columns. Each row is a song with its artists, release date, streams, presence in Spotify, Apple Music and Deezer playlists, chart values for Spotify, Apple Music, Deezer and Shazam, and audio features (BPM, key, mode, danceability, valence, energy, acousticness, instrumentalness, liveness and speechiness).

2.2 Cleaning steps in Power Query

Issue found	Action taken	Result
One row had text instead of a number in streams (Love Grows (Where My Rosemary Goes))	Changed streams to Whole Number and removed the error row	952 rows remain
Numbers with commas in in_deezer_playlists and in_shazam_charts	Confirmed both columns as Whole Number after import	0% errors
50 empty values in in_shazam_charts	Replaced null with 0	Assumption: not on Shazam charts
95 empty values in key	Replaced null with Unknown	Key is not central to the analysis
Multiple artists in one field	Created lead_artist (first named artist)	458 lead artists

2.3 Calculated columns

Column	Logic	Purpose
lead_artist	Text before the first comma in artist(s)_name	Artist-level analysis
streams_millions	streams divided by 1,000,000	Readable axes
release_date	Date built from year, month and day	Time analysis
total_playlists	Spotify + Apple + Deezer playlists	Combined exposure
platforms_charted	Count (0 to 4) of platforms where the chart value is above 0	Cross-platform presence
Playlist Group	Buckets: under 1K, 1K to 3K, 3K to 10K, 10K and above	Compare exposure levels
streams_per_playlist	streams divided by total_playlists	Streams relative to exposure
outlier_type	Percentile rules on streams_per_playlist, streams and playlists	Identify outliers

3. Methodology and rationale

The report is organised as seven Power BI pages: Overview, Audio Features, Cross-Platform, Playlist vs Streams, Chart vs Streams, Outliers and Artist Pareto. Slicers for release year, mode and key make the report interactive, and KPI cards show the headline numbers. DAX measures (for example Total Streams, Avg Streams per Song, Streams per Playlist and a Pearson correlation measure) recalculate with every filter.

Task	Visuals used	Why suitable
Audio features	Column chart of averages, scatter with trend line, BPM histogram, key column chart	Averages summarise the profile; scatter shows relationships; histogram shows spread
Cross-platform	Column charts by platform and by platforms_charted, ranked table	Compares platforms side by side and lists the strongest songs
Playlist vs streams	Scatter with trend line, average streams by playlist group, correlation card	Scatter shows association and unusual points; groups make the pattern easy to read
Chart vs streams	Charted vs not charted comparison, scatters, top 10 table	Compares streams between songs with and without chart presence
Outliers	Coloured scatter, slicer with detail table, comparison matrix	Highlights unusual songs and compares their characteristics
Artist concentration	Ranking, cumulative percentage, Pareto-style chart	Shows how much of total streams the top artists hold

Correlation values quoted below were checked with an independent calculation on the CSV. Playlist correlations are Pearson; chart correlations are rank-based (Spearman) because chart values are skewed.

4. Analysis and findings

4.1 Overview page

The Overview page shows the KPI cards and slicers. For the full dataset the values are: **952 songs**, **458 lead artists**, **about 489.5 billion total streams** and **about 514 million average streams per song** (median about 290.5 million). The gap between the mean and the median shows that a small number of very popular songs pull the average up. Mode is split into 549 Major and 403 Minor songs.

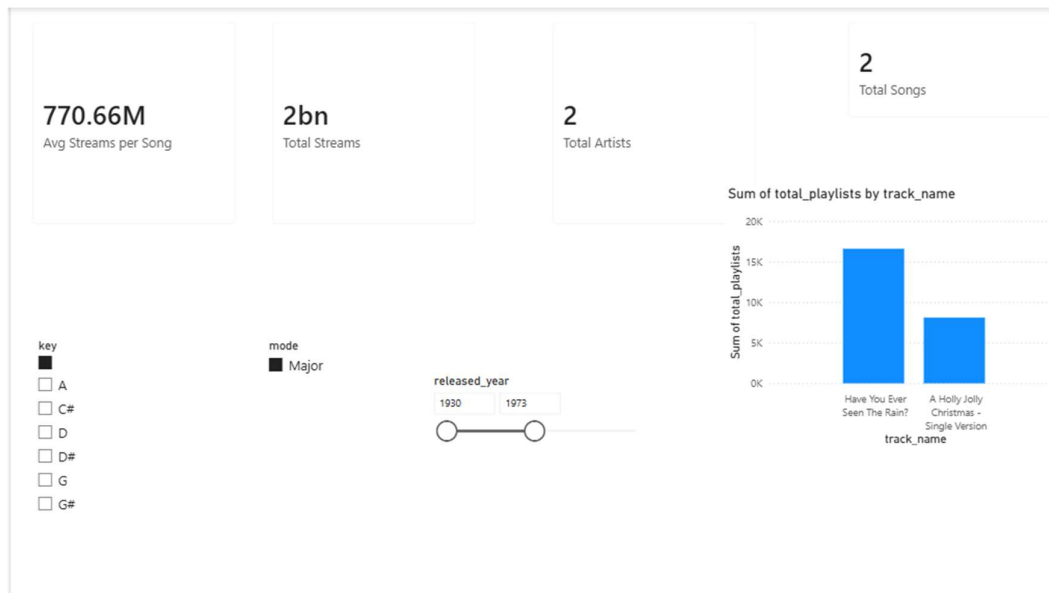


Figure 1: Overview page (KPI cards and slicers)

4.2 Task 1: Music analysis (audio features)

Evidence from the data

- Average values: danceability 67, energy 64, valence 51, acousticness 27, liveness 18, speechiness 10, instrumentalness 2 and BPM about 123.
- About 91% of songs have an instrumentalness of 0, so vocal-led songs dominate.
- Danceability and valence have a correlation of about +0.41. Danceability and acousticness about -0.24. Energy and acousticness about -0.58. Danceability and BPM about -0.15.
- The BPM histogram shows most songs between about 90 and 140 BPM. C# is the most common key (120 songs).

Interpretation

Popular songs in this dataset are typically danceable, energetic and vocal-led, with a mid-tempo pulse. Energetic songs tend to be less acoustic. These are associations between features and do not show that a feature causes popularity.

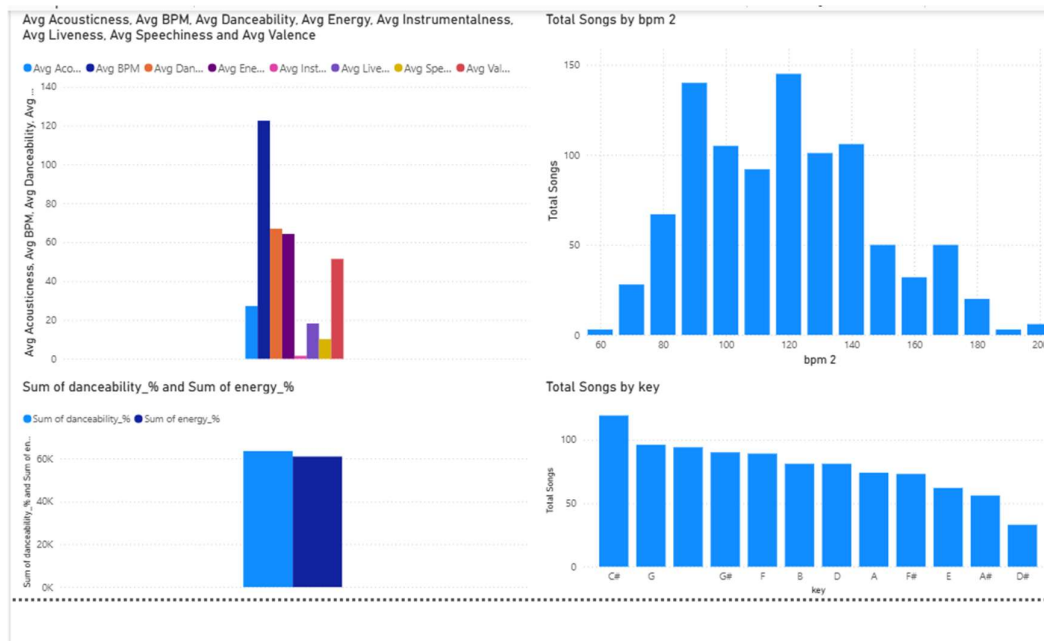


Figure 2: Audio Features page

4.3 Task 2: Cross-platform presence

Evidence from the data

- Songs by number of platform charts: 0 platforms = 55, 1 = 210, 2 = 187, 3 = 229, 4 = 271. So **271 songs (about 28%) chart on all four platforms.**
- Average streams by number of platform charts (millions): 0 = 203, 1 = 330, 2 = 552, 3 = 651, 4 = 579.
- Average playlists per song: Spotify about 5,203, Deezer about 386, Apple Music about 68. These are on different scales, so the comparison shows presence rather than platform size.

Interpretation

Songs with wider chart presence have clearly higher average streams than songs charting on none or one platform. The dip from 3 to 4 platforms shows that more platforms does not always mean more streams. Already-popular songs may simply chart everywhere, so this is an association, not proof that platform presence causes streams.

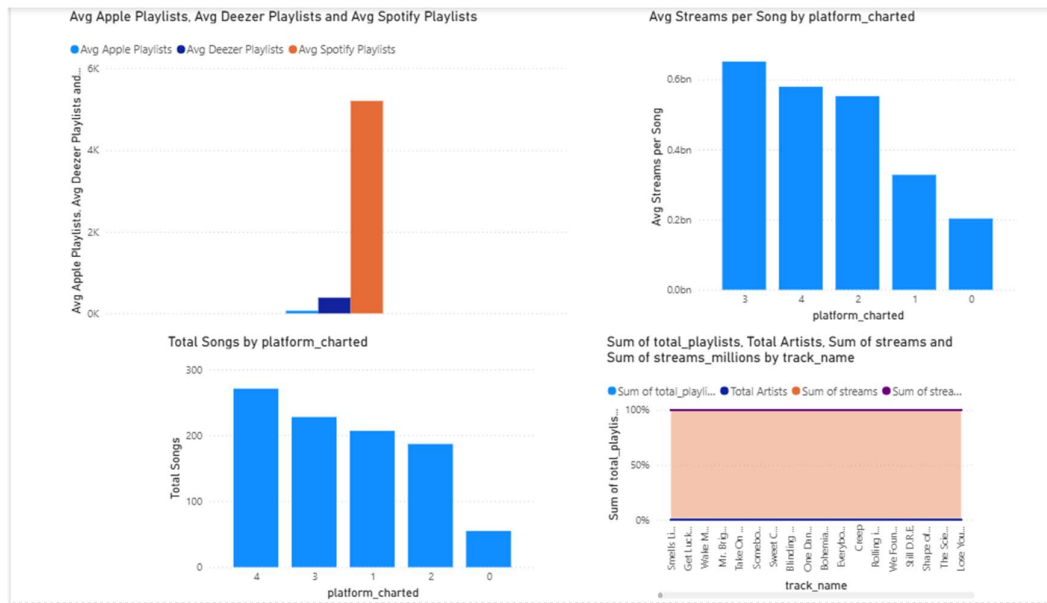


Figure 3: Cross-Platform page

4.4 Task 3: Playlist exposure and streams

Evidence from the data

- Streams and total playlists have a correlation of about **0.78**. Spotify playlists alone about 0.79, Apple Music playlists about 0.77 and Deezer playlists about 0.60.
- Average streams rise steadily across the playlist groups. The very high group (10K and above) is far above the low group.
- The strongest songs lead on both measures. For example, Blinding Lights has about 3.7 billion streams and about 43.9 thousand Spotify playlists.

Interpretation

Songs with greater playlist exposure tend to have higher streams. The link may run both ways: playlists help listeners discover songs, and already-popular songs are added to more playlists. The data cannot separate these two effects.

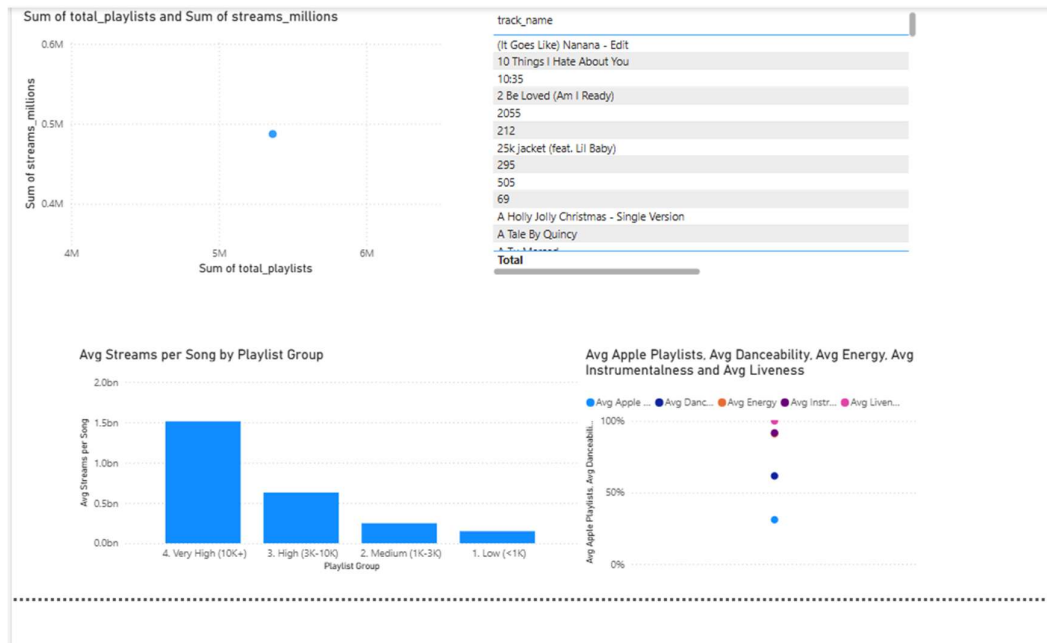


Figure 4: Playlist vs Streams page

4.5 Task 4: Chart position and streams

Assumption: the chart columns behave like a chart presence score where a higher value is stronger and 0 means not charted. This is based on the data: Seven has the highest Spotify value (147) and As It Was has 130.

Evidence from the data

- Average streams for charted vs not charted songs (millions): Spotify 621 vs 369, Apple Music 524 vs 426, Deezer 655 vs 414, Shazam 543 vs 473.
- Correlations with streams are weak to moderate: Spotify about 0.26, Apple Music about 0.33, Deezer about 0.23 and Shazam about 0.12.

Interpretation

Songs with chart presence stream more on average, but chart values explain far less about streams than playlist exposure does. Shazam shows the smallest difference. This is an association only.

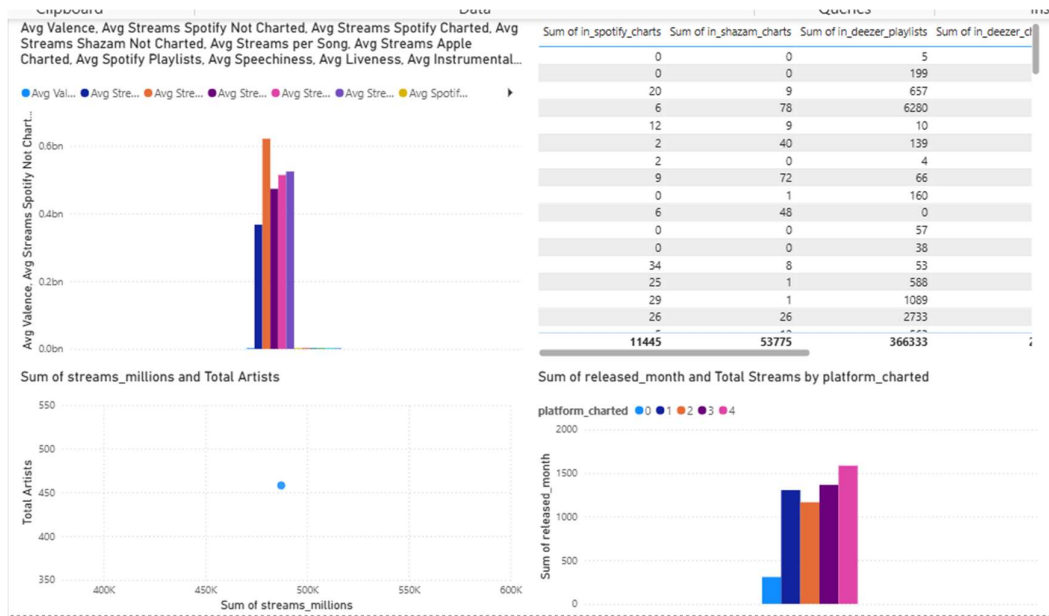


Figure 5: Chart vs Streams page

4.6 Task 5: Outlier and exceptional songs

Two groups were defined using percentiles of streams per playlist. **Group A (high streams, low exposure)**: streams per playlist in the top 10% and streams above the median. **Group B (high exposure, low streams)**: streams per playlist in the bottom 10% and playlists above the median. These thresholds are our own choice.

Evidence from the data

- Group A has 39 songs: about 510 million average streams with only about 960 playlists on average. Median release year is about 2022. Examples: Enemy (from Arcane), Dream On (0 chart platforms, about 839 million streams), Excuses, Kesariya.
- Group B has 73 songs: about 20,400 playlists on average but relatively low streams per playlist. Median release year is about 2008. Examples: Get Lucky, The Next Episode, Christmas (Baby Please Come Home).

Possible interpretation (not proven by the data)

Group A songs may be driven by film or TV use, search or viral sharing rather than playlists. Group B songs look like older or seasonal catalogue tracks that stay on many playlists but earn fewer streams per playlist. Additional data such as traffic sources would be needed to test these ideas.



Figure 6: Outliers page

4.7 Task 6: Artist concentration and 80/20 analysis

Streams were grouped by lead artist (458 artists) and ranked. The values below come from the dataset and represent the Pareto analysis of cumulative percentage contribution.

Measure	Value
Total lead artists	458
Top 1 artist (The Weeknd) share of streams	about 4.4% (about 21.5 billion streams)
Top 5 artists share	about 15.8%
Top 10 artists share	about 24.3%
Top 10% of artists (45) share	about 52.8%
Top 20% of artists (91) share	about 69.0%
Artists needed to reach 80% of streams	about 141 (about 31% of artists)
Artists with only one song in the dataset	about 67%

Top artists by streams: The Weeknd (about 21.5B), Bad Bunny (15.4B), Ed Sheeran (14.6B), Taylor Swift (14.4B) and Harry Styles (11.6B). Taylor Swift has the most songs in the dataset (36).

Interpretation

Streams are concentrated but do not follow a strict 80/20 rule: the top 20% of artists hold about 69% of streams, and about 31% of artists are needed to reach 80%. Success is skewed towards a few major artists, while most artists appear with a single song. For the industry this means a reliance on a small group of superstars, but also a long tail where individual hits can still reach large audiences.

[Figure 7: Artist Pareto page screenshot to be inserted here once the page is built.]

5. Assumptions and limitations

- The row with corrupted streams was removed, leaving 952 songs.
- Empty in_shazam_charts values were treated as 0 (not charted). Empty key values were labelled Unknown.
- lead_artist receives full credit for a song's streams, which avoids double counting but ignores featured artists.
- Chart columns were treated as a presence score with 0 meaning not charted.
- total_playlists adds playlists from platforms with different sizes.
- Outlier thresholds (10th and 90th percentile with median filters) are analyst choices.
- The dataset is a snapshot of popular songs, so it may not represent all music. All relationships are correlations, not causes.

6. Implications and conclusion

- **Playlists matter most:** playlist exposure shows the strongest association with streams (about 0.78), so playlist strategy is a key area for labels and artists.
- **Charts are weaker signals:** chart presence relates to streams, but much less strongly, and Shazam is the weakest.
- **Platform breadth helps, with limits:** songs on 3 or more platform charts stream far more than those on 0 or 1.
- **Sound profile:** popular songs are mostly danceable, energetic, vocal-led and mid-tempo.
- **Exceptions exist:** some songs stream heavily with few playlists, and some are widely playlisted with modest streams, so playlist counts alone should not be used to judge a song.
- **Concentration:** a small group of artists holds a large share of streams, though less than a strict 80/20 split.

Overall, the analysis shows that visibility across playlists and platforms goes together with higher streaming, and that a few artists dominate total streams. These are associations found in the data and should be tested further before drawing causal conclusions.